# Haotian Zhang

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## Education

**University of Southern California** Starting Spring 2027
Incoming undergraduate in Applied and Computational Mathematics.

**American University of Paris** August - December 2026
Undergraduate coursework; Fall 2026 semester.

## Research Interests

Video reasoning; world models; applied mathematics.

I explore how models reason through visual representations, predict a changing world, and maintain a coherent understanding of objects and physical interactions.

## Selected Honors and Awards

**Hugging Face #1 Paper of the Week** September 2026
For *Training Object Permanence in World Models*.

**United States of America Mathematical Olympiad (USAMO)** 2024
Qualifier; American Invitational Mathematics Examination (AIME): 13/15.

**American Mathematics Competition (AMC 12)** 2024
Global Top 1%.

**S.-T. Yau High School Science Award - Mathematics** 2024
Second Prize (Top 8), Southern Region Finals.

**Berkeley Math Tournament** 2023
National High Distinction (individual); national Top 2 overall and Top 3 in the Guts round.

**High School Mathematical Contest in Modeling (HiMCM)** 2024
Honorable Mention.

## Publications

### Preprint

**Training Object Permanence in World Models**

**Haotian Zhang**, Fengyuan Yu, Dezhi Luo, Haoran Sun, Zehong Zhao, Qingying Gao, ..., Yilun Du, Hokin Deng.

*arXiv:2609.28654, 2026.* Co-first author.

Paper | Project | Code

## Research Experience

**Hippocampal, Inc.** June 2026 - Present
*Research Intern*

**Training Object Permanence in World Models (WROP)**

- Designed 150 Blender-based 3D task generators to study object permanence and solidity, supporting the project's 1.5-million-sample training corpus.
- Contributed to training a 16B-parameter video model, human evaluation, and manuscript preparation.
- Co-first authored the resulting preprint, recognized as Hugging Face's #1 Paper of the Week.

## Research Experience (continued)

**University of California, Santa Cruz** June - August 2025
*Research Intern, Summer Internship Program (SIP)*

- Developed a generative-AI tool to scaffold research-paper reading through simulated reader-AI dialogue and iterative refinement of system prompts.
- Designed and evaluated prompts to help AI systems explain complex academic texts.
- Evaluated NotebookLM's Audio Overview for vocal tone, pacing, and content accuracy against professional podcast standards; explored prompt strategies for more natural narration.

**Cambridge Centre for International Research (CCIR)** June - September 2025
*Research Student*

- Completed coursework in Applied Linguistics & Big Data.
- Authored a research paper on the social functions of multilingualism along the ninth- and tenth-century Silk Road, combining linguistic theory with data-informed analysis.